

Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 9 - Electron Spin

Solution. 9.2

- (a) For a particle with $s = 3/2$ there are four possible values of S_z and they are

$$\frac{3}{2}\hbar, \quad \frac{1}{2}\hbar, \quad -\frac{1}{2}\hbar, \quad -\frac{3}{2}\hbar$$

- (b) Below we show the requested vector model diagram

- (c) The minimum possible angle between $\mathbf{S}$ and the z -axis is given by

$$\cos \theta = \frac{3\hbar/2}{\sqrt{15}\hbar/2} = \frac{3}{\sqrt{15}}$$

Which implies that $\theta \approx 39.23^\circ$

□

Solution. 9.3

- (a) For a particle with $s = 1$ there are three possible values of S_z and they are

$$\hbar, \quad 0, \quad -\hbar$$

- (b) Below we show the requested vector model diagram

- (c) The minimum possible angle between $\mathbf{S}$ and the z -axis is given by

$$\cos \theta = \frac{\hbar}{\hbar} = 1$$

Which implies that $\theta = 0^\circ$

□

Solution. 9.6

(a) For an s electron we have that $L = \sqrt{0(0+1)}\hbar = 0$ so

$$\frac{S}{L} = \frac{\sqrt{3}\hbar/2}{0} = \infty$$

(b) For a p electron we have that $L = \sqrt{1(1+1)}\hbar = \sqrt{2}\hbar$ so

$$\frac{S}{L} = \frac{\sqrt{3}\hbar/2}{\sqrt{2}\hbar} = \frac{\sqrt{6}}{4} = 0.612$$

(c) For a d electron we have that $L = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar$ so

$$\frac{S}{L} = \frac{\sqrt{3}\hbar/2}{\sqrt{6}\hbar} = \frac{\sqrt{2}}{4} = 0.353$$

(d) For an f electron we have that $L = \sqrt{3(3+1)}\hbar = \sqrt{12}\hbar$ so

$$\frac{S}{L} = \frac{\sqrt{3}\hbar/2}{\sqrt{12}\hbar} = \frac{1}{4} = 0.25$$

□

Solution. 9.8 If we consider the electron to be a spinning ball of matter, considering the velocity at the equator v and the upper bound of the radius of the electron r , then the angular momentum is

$$L = m_e v r$$

Since we know that $L = \sqrt{3}\hbar/2$ then we can determine the velocity of rotation to be

$$v = \frac{L}{m_e r} = \frac{\sqrt{3}\hbar/2}{9.109 \times 10^{-31} \cdot 10^{-18}} = 1.003 \times 10^{14} \text{ m/s}$$

Then we see that

$$\frac{v}{c} = \frac{1.003 \times 10^{14}}{2.998 \times 10^8} = 334556.37$$

Implying that the electron is spinning 334556.37 times faster than the speed of light, which is not possible according to relativity. $\square$

Solution. 9.9 Let a current of 0.4 A flow around a single circular loop of radius 1 cm .

(a) The magnitude of the magnetic moment, μ , is

$$\mu = iA = i\pi r^2 = 1.256\text{ Acm}^2 = 0.0001256\text{ Am}^2$$

(b) Let $\boldsymbol{\mu}$ be perpendicular to $\boldsymbol{B}$ where $B = 1.5\text{ T}$ then the torque on the loop has a magnitude

$$\Gamma = \mu B \sin \theta = \mu B = 0.0001256\text{ Am}^2 \cdot 1.5\text{ T} = 0.0001883\text{ Nm}$$

(c) When $\boldsymbol{\mu}$ is parallel to $\boldsymbol{B}$ the energy is

$$U_p = -\mu B \cos(0) = -\mu B = -0.0001883\text{ J}$$

And when they are antiparallel the energy is

$$U_{ap} = -\mu B \cos(\pi) = \mu B = 0.0001883\text{ J}$$

So the difference in energy is

$$|U_p - U_{ap}| = |-0.0001883\text{ J} - (0.0001883\text{ J})| = 0.0003766\text{ J}$$

□

Solution. 9.11 Given that a typical atomic magnetic moment is of order 10^{-23} Am^2 assuming a single circular loop of radius 0.1 nm then i is

$$i = \frac{\mu}{A} = \frac{\mu}{\pi r^2} = \frac{10^{-23}}{\pi(0.1 \times 10^{-9})^2} = 0.000318 \text{ A}$$

□

Solution. 9.12 The magnetic energy due to the orbital magnetic moment in the direction of the z -axis is

$$U_z = -\mu_z B = -\frac{e}{2m_e} \hbar \cdot B$$

Then for a $B = 10 \text{ T}$ we have that

$$\begin{aligned} U_z &= -\frac{1.602 \times 10^{-19} \text{ C}}{2 \cdot 9.109 \times 10^{-31} \text{ kg}} \cdot 1.055 \times 10^{-34} \frac{\text{m}^2 \text{kg}}{\text{s}} \cdot 10 \text{ T} \\ &= -9.277 \times 10^{-23} \text{ J} = -5.79 \times 10^{-4} \text{ eV} \end{aligned}$$

□

Solution. 9.15 Let a helium atom in an energy level with one electron occupying an s state ($l = 0$) and the other in an f state ($l = 3$). The two electron spins are antiparallel so that the spin magnetic moments cancel. The atom is placed in a magnetic field $B = 0.8 \text{ T}$.

(a) Below, on the right, we show a sketch of the resulting splitting of the original energy level.

(b) The energy difference between adjacent levels of the resulting multiplet is given by

$$\mu_B B = 5.79 \times 10^{-5} \text{ eV/T} \cdot 0.8 \text{ T} = 4.63 \times 10^{-5} \text{ eV}$$

Assuming the magnetic field is in the direction of L_z .

□

Solution. 9.16

- (a) The B field will not affect the $1s$ state since $l = 0$ for this case and hence the only value m can take is $m = 0$.
 For the state $2p$ we have that $l = 1$ so m can take the values $-1, 0$ and 1 . Then the magnetic field B will split this energy level into three different ones.

- (b) When B is switched on there are three spectral lines due to the split of the energy level E_2 associated to the state $2p$.
 (c) We know that the energy difference between the state $1s$ and $2p$ is

$$E_2 - E_1 = -3.4\text{eV} + 13.6\text{eV} = 10.2\text{ eV}$$

So the frequency of the photon emitted when the electron drops from the state $2p$ to $1s$ is

$$f_0 = \frac{E_2 - E_1}{h} = \frac{10.2\text{ eV}}{4.135 \times 10^{-15}\text{ eVs}} = 2.466 \times 10^{15}\text{ Hz}$$

And the frequency difference due to the magnetic field, Δf , is

$$\Delta f = \frac{\mu_B B}{h} = \frac{5.79 \times 10^{-5}\text{ eV/T} \cdot 1.5\text{ T}}{4.135 \times 10^{-15}\text{ eVs}} = 2.1 \times 10^{10}\text{ Hz}$$

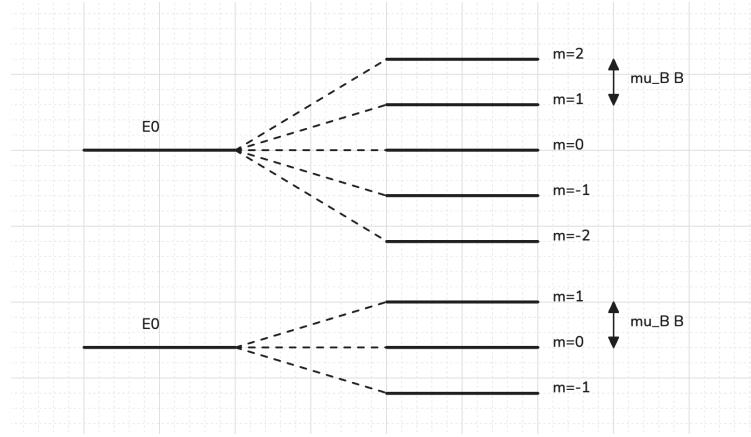
Then the fractional separation between adjacent lines is

$$\frac{\Delta f}{f_0} = \frac{2.1 \times 10^{10}\text{ Hz}}{2.466 \times 10^{15}\text{ Hz}} = 8.515 \times 10^{-6}$$

□

Solution. 9.17

- (a) In the top part of the sketch we show the split of the electron's energy in the d state and in the lower part the split of the electron's energy in the p state due to the magnetic field.



- (b) The photon energy is determined by

$$\begin{aligned} E_i - E_f &= (E_d + m_i \mu_B B) - (E_p + m_f \mu_B B) \\ &= E_d - E_p + \mu_B B (m_i - m_f) \end{aligned}$$

So the number of possible photon energies is given by the number of different $(m_i - m_f)$ we can have.

If $m_i = 2$ the different differences we can have are

$$\begin{aligned} 2 - 1 &= 1 \\ 2 - 0 &= 2 \\ 2 - (-1) &= 3 \end{aligned}$$

If $m_i = 1$

$$\begin{aligned} 1 - 1 &= 0 \\ 1 - 0 &= 1 \text{ (repeated)} \\ 1 - (-1) &= 2 \text{ (repeated)} \end{aligned}$$

If $m_i = 0$

$$\begin{aligned} 0 - 1 &= -1 \\ 0 - 0 &= 0 \text{ (repeated)} \\ 0 - (-1) &= 1 \text{ (repeated)} \end{aligned}$$

If $m_i = -1$

$$\begin{aligned} -1 - 1 &= -2 \\ -1 - 0 &= -1 \text{ (repeated)} \\ -1 - (-1) &= 0 \text{ (repeated)} \end{aligned}$$

And if $m_i = -2$

$$-2 - 1 = -3$$

$$-2 - 0 = -2 \text{ (repeated)}$$

$$-2 - (-1) = -1 \text{ (repeated)}$$

Therefore the possible differences are

$$(m_i - m_f) \in \{-3, -2, -1, 0, 1, 2, 3\}$$

So there are 7 possible photon energies.

(c) Because the difference $(m_i - m_f)$ can only take the values $-1, 0$ or 1 then because of what we saw in part (b) we get that there are only three possible photon energies. $\square$

Solution. 9.18 Let $\boldsymbol{\mu} = -e(\mathbf{L} + 2\mathbf{S})/2m_e$ be the electron's total magnetic moment.

- (a) If $l = 0$ then we get that $\mathbf{L} = 0$ and hence the total magnetic moment of the electron becomes

$$\boldsymbol{\mu} = -\frac{e}{m_e}\mathbf{S}$$

So the total magnetic moment is reduced to the spin magnetic moment of the electron.

Since the component S_z is given by $S_z = m_s\hbar$ and m_s can be $1/2$ or $-1/2$ for the electron then μ_z can have two values

$$\mu_z = -\frac{e}{m_e}S_z = \begin{cases} -\frac{e\hbar}{2m_e} & m_s = \frac{1}{2} \\ +\frac{e\hbar}{2m_e} & m_s = -\frac{1}{2} \end{cases}$$

- (b) A hypothetical spinless electron with $l = 1$ will have a magnetic moment component μ_z given by

$$\mu_z = -\frac{e}{2m_e}L_z = -\frac{e}{2m_e}m\hbar$$

Where $m = -1, 0, 1$ hence

$$\mu_z = -\frac{e}{2m_e}m\hbar = \begin{cases} -\frac{e\hbar}{2m_e} & m = 1 \\ 0 & m = 0 \\ +\frac{e\hbar}{2m_e} & m = -1 \end{cases}$$

So compared to part (a) they would have the same magnetic moment component μ_z .

□

Solution. 9.19 Let a hydrogen atom on its ground level placed in a magnetic field of 0.7 T along the z axis.

- (a) The energy difference between the spin-up and spin-down states is given by

$$\Delta E = 2\mu_B B = 2 \cdot 5.79 \times 10^{-5} \cdot 0.7 = 8.106 \times 10^{-5} \text{ eV}$$

- (b) To excite the atom from the lower to the upper state we need to send a photon with an energy of $8.106 \times 10^{-5} \text{ eV}$ since that's the energy separation between the levels. Hence it would need to have a wavelength given by

$$\lambda = \frac{hc}{\Delta E} = \frac{1240 \text{ eV} \cdot \text{nm}}{8.106 \times 10^{-5} \text{ eV}} = 15297310.63 \text{ nm} = 1.529 \text{ cm}$$

This radiation falls under the microwave radiation.

□

Solution. 9.20 Let us consider a hydrogen atom in the 3d state with $L_z = 2\hbar$ and $S_z = \frac{1}{2}\hbar$. If we place it in a magnetic field $B = 0.6 \text{ T}$ along the z axis its energy will change by

$$\Delta E = -\boldsymbol{\mu}_{tot} \cdot \mathbf{B} = -\frac{e}{2m_e}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B} = -\frac{e}{2m_e}(L_z + 2S_z)B$$

Hence

$$\begin{aligned}\Delta E &= -\frac{e}{2m_e}(2\hbar + \hbar) \cdot 0.6 \\ &= -\frac{e\hbar}{2m_e} \cdot 1.8 \\ &= -\mu_B \cdot 1.8 \\ &= -5.79 \times 10^{-5} \cdot 1.8 \\ &= -1.0422 \times 10^{-4} \text{ eV}\end{aligned}$$

□

Solution. 9.21

- (a) We know that the B field at the center of a circular current loop, i , of radius r is known to be

$$B = \frac{\mu_0 i}{2r}$$

Since the current produced by the orbiting proton is

$$i = \frac{ev}{2\pi r}$$

Then

$$B = \frac{\mu_0 ev}{4\pi r^2}$$

Multiplying and dividing the expression by $m_e r$ where m_e is the mass of the electron, we get that

$$B = \frac{\mu_0 e m_e v r}{4\pi m_e r^3} = \frac{\mu_0}{4\pi} \frac{eL}{m_e r^3}$$

Where we used that $L = m_e v r$ is the electron's orbital angular momentum.

- (b) Let $L = 2\hbar$ and $r = 4a_B$ then B is given by

$$\begin{aligned} B &= \frac{\mu_0}{128\pi} \frac{e\hbar}{m_e a_B^3} \\ &= \frac{4\pi \times 10^{-7} \text{ Tm/A}}{128\pi} \frac{1.602 \times 10^{-19} \text{ C} \cdot 1.054 \times 10^{-34} \text{ Js}}{9.109 \times 10^{-31} \text{ kg} \cdot (5.29 \times 10^{-11} \text{ m})^3} \\ &= 0.39 \text{ T} \end{aligned}$$

Hence, the separation of the two $2p$ levels is

$$2\mu_B B = 2 \cdot 5.79 \times 10^{-5} \text{ eV/T} \cdot 0.39 \text{ T} = 4.51 \times 10^{-5} \text{ eV}$$

□

Solution. 9.22

(a) The separation of the two $2p$ levels ΔE_{FS} can be written as

$$\begin{aligned}
 \Delta E_{FS} &= 2\mu_B B \\
 &= 2 \frac{e\hbar}{2m_e} \frac{\mu_0}{4\pi} \frac{2e\hbar}{64m_e a_B^3} \\
 &= 2 \frac{e\hbar}{2m_e} \frac{\mu_0 \varepsilon_0}{4\pi \varepsilon_0} \frac{2e\hbar (ke^2 m_e)^3}{64m_e (\hbar^2)^3} \\
 &= \frac{e\hbar}{m_e} \frac{k}{c^2} \frac{e\hbar (ke^2 m_e)^3}{32m_e \hbar^6} \\
 &= \frac{\hbar^2 (ke^2)^4 m_e^3}{32m_e^2 \hbar^6 c^2} \\
 &= \frac{m_e (ke^2)^4}{32\hbar^4 c^2}
 \end{aligned}$$

Where we replaced the values of a_B and μ_B .

(b) We know that

$$E_R = \frac{m_e (ke^2)^2}{2\hbar^2}$$

Then we can write ΔE_{FS} as

$$\Delta E_{FS} = \frac{m_e (ke^2)^2}{2\hbar^2} \frac{(ke^2)^2}{16\hbar^2 c^2} = E_R \frac{(ke^2)^2}{16\hbar^2 c^2}$$

Defining $\alpha = ke^2/\hbar c$, we see that

$$\Delta E_{FS} = \frac{\alpha^2 E_R}{16}$$

(c) We see that

$$\alpha = \frac{ke^2}{\hbar c} = \frac{8.98 \times 10^9 \cdot (1.602 \times 10^{-19})^2}{1.054 \times 10^{-34} \cdot 2.9979 \times 10^8} = 0.007299 \approx \frac{1}{137}$$

□

Solution. 9.23

(a) The relativistic kinetic energy expression for the electron is given by

$$K = mc^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)$$

Applying the binomial series we get that

$$\begin{aligned} K &= mc^2 \left[\left(1 + \frac{1}{2} \frac{v^2}{c^2} + \frac{-1/2(-1/2-1)}{2} \frac{v^4}{c^4} \right) - 1 \right] \\ &= mc^2 \left(\frac{1}{2} \frac{v^2}{c^2} + \frac{3/4}{2} \frac{v^4}{c^4} \right) \\ &= \frac{1}{2} mv^2 + \frac{3}{8} \frac{mv^4}{c^2} \end{aligned}$$

(b) Replacing $v = \alpha c$, the relativistic correction gives us

$$\Delta E_{rel} = \frac{3}{8} \frac{mv^4}{c^2} = \frac{3}{8} \frac{m\alpha^4 c^4}{c^2} = \frac{3}{8} \alpha^2 \frac{m(ke^2)^2 c^2}{\hbar^2 c^2} = \frac{3}{4} \alpha^2 \frac{m(ke^2)^2}{2\hbar^2} = \frac{3\alpha^2 E_R}{4}$$

Where we replaced $\alpha = ke^2/\hbar c$.

□

Solution. 9.24

(a) Let us compute the value of the nuclear magneton as follows

$$\mu_N = \frac{e\hbar}{2m_p} = \frac{1.602 \times 10^{-19} \cdot 1.054 \times 10^{-34}}{2 \cdot 1.672 \times 10^{-27}} = 5.049 \times 10^{-27} \text{ J/T}$$

or in eV/T we get that

$$\begin{aligned}\mu_N &= 5.049 \times 10^{-27} \text{ J/T} \cdot 6.242 \times 10^{18} \text{ eV/J} \\ &= 3.151 \times 10^{-8} \text{ eV/T}\end{aligned}$$

This value is about 2000 times smaller compared to Bohr magneton $\mu_B = 5.79 \times 10^{-5} \text{ eV/T}$ because the proton is much heavier.

(b) The energy-level separation is given by

$$\begin{aligned}\Delta E &= 2\mu_p B \\ &= 2 \cdot 2.7\mu_N B \\ &= 2 \cdot 2.7 \cdot 3.151 \times 10^{-8} \text{ eV/T} \cdot 1 \text{ T} \\ &= 1.7014 \times 10^{-7} \text{ eV}\end{aligned}$$

The thermal energy at room temperature is

$$k_B T = 1.380 \times 10^{-23} \cdot 298 = 4.112 \times 10^{-21} \text{ J}$$

Or in eV

$$\begin{aligned}k_B T &= 4.112 \times 10^{-21} \text{ J} \cdot 6.242 \times 10^{18} \text{ eV/J} \\ &= 0.0256 \text{ eV}\end{aligned}$$

The ratio between the thermal energy and the energy-level separation is

$$\frac{k_B T}{\Delta E} = \frac{0.0256 \text{ eV}}{1.7014 \times 10^{-7} \text{ eV}} = 150464.32$$

So the thermal energy is about 15000 times bigger than the energy-level separation.

□

Solution. 9.25 Given that the energy-level separation is

$$\Delta E = 1.7014 \times 10^{-7} \text{ eV}$$

This energy-level separation is associated with a frequency

$$\hbar 2\pi f = \Delta E$$

$$f = \frac{\Delta E}{2\pi \hbar}$$

$$f = \frac{1.7014 \times 10^{-7} \text{ eV}}{2\pi \cdot 1.054 \times 10^{-34} \text{ Js} \cdot 6.242 \times 10^{18} \text{ eV/J}}$$

$$f = 41158750.77 \text{ Hz} \approx 41.15 \text{ MHz}$$

□

Solution. 9.27

Let $\mathbf{\Gamma} = \boldsymbol{\mu} \times \mathbf{B}$ be the torque on a moment in a field. We know also, that the angular momentum is related to the moment $\boldsymbol{\mu}$ by $\boldsymbol{\mu} = \gamma \mathbf{L}$ and that the torque is related to the angular momentum by $\mathbf{\Gamma} = d\mathbf{L}/dt$.

Classically, precession exists when a torque changes the angular momentum direction. In this cases, we notice that $\mathbf{\Gamma}$ is perpendicular to $\boldsymbol{\mu}$ and $\mathbf{B}$ and the angular momentum $\mathbf{L}$ is proportional to $\boldsymbol{\mu}$ so the torque is also perpendicular to $\mathbf{L}$, since they are related by $\mathbf{\Gamma} = d\mathbf{L}/dt$ then the change in the angular momentum is perpendicular to $\mathbf{L}$ hence $\mathbf{L}$ changes direction, and therefore precession exists and since $\mathbf{L}$ is proportional to $\boldsymbol{\mu}$ then it also precesses about the field direction.

If we compare the equation we have

$$\begin{aligned}\mathbf{\Gamma} &= \boldsymbol{\mu} \times \mathbf{B} \\ \frac{d\mathbf{L}}{dt} &= \gamma \mathbf{L} \times \mathbf{B}\end{aligned}$$

To the general equation for a spinning top

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\omega} \times \mathbf{L}$$

We notice that the magnitude of the angular velocity in our case must be

$$\omega = \gamma B$$

□